Veille informatique quantique - 2026-10-16

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. Energetics of Fault-Tolerant Spin-Qubit Quantum Computation

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 87/100 | nouveauté 50/100 | importance 50/100 | score 67/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.22199v1 Announce Type: new Abstract: Semiconductor spin qubits offer a promising route to compact and scalable quantum computers, combining long coherence times in isotopically purified silicon with compatibility with semiconductor manufacturing. Beyond computational speedup, their energy efficiency is an important consideration for future large-scale systems. We evaluate the power and energy requirements of a Loss-DiVincenzo spin-qubit quantum computer by analyzing the Quantum Fouri

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, quantum computer, qubit, quantum error correction.

2. Perfect and peak state transfer on IBM quantum hardware

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.20855v1 Announce Type: new Abstract: We run continuous-time quantum walks with perfect and peak state transfer on a present-day superconducting quantum processor, IBM Quantum's 156-qubit Heron device ibm_kingston, deliberately without error correction or error mitigation. Perfect state transfer and its relaxations have been studied extensively in algebraic graph theory, where one can prove exactly which graphs move a quantum state from one vertex to another with probability one (or

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, quantum processor, IBM Quantum.

3. Superconducting circuits host two-level systems with lifetimes exceeding one second

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.21853v1 Announce Type: new Abstract: Understanding the microscopic environment of superconducting qubits is a central challenge for quantum technologies. Parasitic two-level systems (TLSs) of various microscopic origins are ubiquitous and limit qubit performance through decoherence, frequency drifts, and leakage. Recently, long-lived TLSs with lifetimes exceeding those of state-of-the-art qubits have been shown to induce correlated qubit dynamics, with direct implications for quantum

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, superconducting qubit, quantum error correction.

4. Programmable all-to-all entangling gates for cavity-coupled atoms

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.22220v1 Announce Type: new Abstract: Fault tolerant quantum computers greatly benefit from having both high fidelity entangling gates and a non-local qubit connectivity. However, most entangling gates for neutral atoms are local, requiring separated atoms to be moved using optical tweezers to yield entanglement between them. Here, we design a high fidelity M\o lmer S\o rensen-like entangling gate between neutral atoms coupled to a photonic cavity, enabling a programmable all-to-all c

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, neutral atoms.

5. Realization Divergence in Multi-Provider Quantum Computing: A Pre-Registered Study of What Survives Measurement

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.21714v1 Announce Type: cross Abstract: Quantum programs pass through compilers, transpilers and vendor services before execution. However, the distance between what was written and what runs is rarely measured. We call this distance realization divergence: the observable difference in how an identical source circuit is realized and behaves across quantum software stacks. We report a hardware study across three vendors and four devices, combining pre-registered predictions with explic

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, IonQ.

6. Jianmu-Q: Bridging Quantum Field Theory and Quantum Hardware with AI Agents

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.21942v1 Announce Type: cross Abstract: Quantum computing offers a promising approach to non-perturbative simulations of quantum field theories, particularly for real-time dynamics and systems affected by the sign problem. Yet translating continuum field theories into quantum computations remains a challenging interdisciplinary task. Here we introduce \texttt{Jianmu-Q}, a multi-agent framework that transforms natural-language descriptions of quantum field theories into quantum simulat

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, IBM Quantum.

7. Error Correction in a Distributed Quantum Computer

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.13065v2 Announce Type: replace Abstract: Building fault-tolerant quantum computers with large numbers of logical qubits requires both scalable hardware architectures and error-correcting codes that make efficient use of physical qubits. Photonic interconnects address both of these challenges by allowing the physical qubits to be distributed across multiple processors while providing the non-local connectivity necessary to implement resource-efficient codes such as high-rate quantum l

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, quantum error correction.

8. Quantum computational advantage in random-circuit sampling on IBM superconducting quantum computers

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.28657v2 Announce Type: replace Abstract: We report forward random-circuit sampling (RCS) on the 120-qubit Nighthawk r2 superconducting processor (ibm_phoenix) with square-lattice connectivity, using 61 qubits, native CZ gates, and the standard cloud execution stack with no benchmark-specific calibration. Two independent fidelity estimators---mirror benchmarking and three- and four-patch cross-entropy benchmarking (XEB)---agree with each other at every measured depth, the mirror from

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, quantum processor.

9. A modular quantum computer based on Cornucopia codes

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.19073v2 Announce Type: replace Abstract: Ultra-high-rate quantum error correction codes store many logical qubits in a single code block with few physical qubits. A fault-tolerant quantum computer, however, also requires a universal set of logical operations on individually addressable logical qubits, and the ancilla overhead of these operations may erode the advantage of the high encoding rate. Here, we propose the Cornucopia architecture, in which a universal set of logical operati

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, quantum error correction.

10. Syndrome-Resolved Decoding of Qubit Readout Errors

Source : arXiv Quantum Physics - publié le 2026-10-16

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.20810v1 Announce Type: new Abstract: Readout errors in superconducting qubits include incorrect state assignment, bit flips within the computational subspace, and leakage to non-computational states. These errors require different mitigation strategies, yet conventional measurement fidelity and repeatability metrics do not resolve their individual contributions. Previous repeated-readout studies have shown that temporal correlations reveal additional information about readout errors.

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, superconducting qubit.

Méthode

Collecte Inoreader/RSS, filtre lexical documenté, puis analyse assistée par IA. Les liens vers les sources permettent la vérification.